GGG-S-113A
May 11, 1992
SUPERSEDING
GGG-S-113
June 10, 1963

FEDERAL SPECIFICATION

SCRAPER, BEARING

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers scrapers for use in scraping bearings on both flat and curved surfaces.

1.1.1 Federal specification coverage. Federal specifications do not cover all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types and classes. Scrapers shall be of the following types and classes as specified (see 6.1):

Type I - Triangle (three square)
Type II - Half round (spoon)
Type III - Flat
  Class 1 - End cutting, carbide tip
  Class 2 - End cutting, steel
Type IV - Double end, steel

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-P-40 - Preservation and Packing of Hand Tools; Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery

Federal Standards:

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and Commercial Item Descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, and other Federal specifications and Commercial Item Descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM):

E 18 - Metallic Materials Rockwell Hardness and Rockwell Superficial Hardness of

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

3.1 Illustrations. Illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of scrapers otherwise conforming to this specification.

3.2 Material. All scraper blades, except type III, class 1, shall be forged of either a high-grade carbon or alloy steel. Type III, class 1 scrapers shall be as hereinafter specified.

3.2.1 Handles. Handles shall be of straight-grained hardwood, free from harmful defects. Moisture content shall not exceed 12 percent. Handles shall be of commercial design and shall be entirely suitable for the purpose intended. Each handle shall be fitted with a metal ferrule of sufficient strength and of the proper design to prevent splitting of the wood when the scraper tang is forced in the handle. Handles shall be securely attached to the tang of the scraper. Unless otherwise specified (see 6.1), handles shall be furnished on all scrapers, except type IV.

3.3 Hardness. Scraper blades, except type III, class 1, shall be hardened to not less than 58 nor more than 64 on the Rockwell C scale. Type III, class 1 blades shall be as hereinafter specified.

3.4 Marking. Scrapers shall be marked with the manufacturer's name or identifying symbol and the state or country of manufacture. The marking shall be engraved, etched, or stamped in such a manner that it will be permanent to the extent that it will remain clear and legible throughout the life of the item.

3.5 Finish.

3.5.1 Blades. Steel scraper blades shall be sharpened and ready for use. The edges shall be free of nicks and rough grind marks. The remainder of the blade shall have either an oil, ground, or enameled finish with no rough or sharp corners.

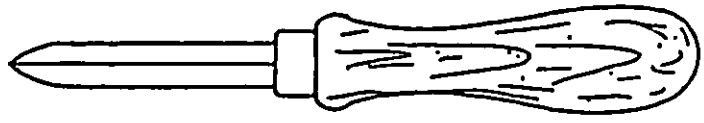
3.5.2 Handle finish. Handles shall be smoothly finished and shall have a protective coating of clear or colored varnish or lacquer.

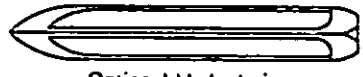
3.6 Type I, triangle (three square). Type I scrapers shall be triangular shaped. The blades shall be hollow ground on all three sides to form the cutting edges. The blade, at the option of the manufacturer, may be relieved not more than one-half of the width of the blade. The scraper shall be straight from the tang, terminating in either a sweeping curve or tapering on all sides at least one-fourth of the blade length to form a point. Type I scrapers shall conform to table I and shall be similar to either option of figure 1.

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TABLE I. Type I, triangle (three square)

Blade length ± 1/2	Dimension across flats ± 1/32	Handle length ± 1/2	Handle diameter ± 1/4
Inches	Inch	Inches	Inches
3	3/8	4	1
4	3/8	4	1
4	7/16	4-1/4	1-1/4
6	7/16	4-1/4	1-1/4
8	7/16	4-1/4	1-1/4
8	5/8	4-1/4	1-1/4





Optional blade design

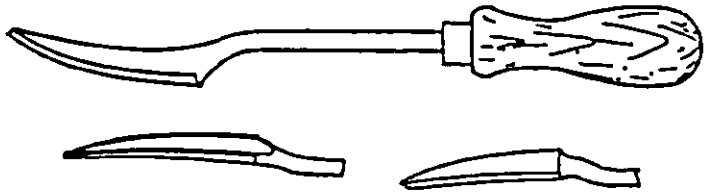
Figure 1. Type I, triangle (three square).

3.7 Type II, half round (spoon). Type II scrapers shall consist of a spoon-shaped blade, a shank, and handle. The blade may be either hollow ground or relieved with not less than 1/16 to 1/8 inch flat between the edge of the blade and the relief. The shank may be round or rectangular with the corners relieved of all burrs. Type II scrapers shall conform to table II and shall be similar to either option of figure 2.

TABLE II. Type II, half round (spoon)

Blade length ± 1/8	Blade width at widest point	Shank length ± 1	Handle length ± 1/2	Handle diameter ± 1/4
Inches	Inch	Inches	Inches	Inches
1-1/2	7/16 ± 1/32	2-3/4	4	1-1/8
2	7/16 ± 1/32	3	4	1-1/8
2-1/2	9/16 ± 1/32	3-3/8	4	1-1/4
3	5/8 ± 1/8	4	4-1/4	1-1/4
3-1/2	5/8 ± 1/8	3	4-3/4	1-3/8
4	11/16 ± 1/8	4	4-3/4	1-3/8
4	13/16 ± 1/8	3	4-3/4	1-3/8

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Optional design

Figure 2. Type II, half round (spoon).

3.8 Type III, flat.

3.8.1 Class 1, end cutting, carbide tip. Class 1 scrapers shall have a carbide tip silver soldered or brazed to a rolled steel shank. The steel shank shall have the corners rounded with not less than 1/16-inch radius. The carbide tip shall be in accordance with the carbide industry chart of grade recommendations for recommendation C-2. Both cutting edgers of the tip shall be sharpened. The steel shank shall be hardened to not less than 28 nor more than 45 of the Rockwell C scale. Class 1 scrapers shall conform to table III and be similar to figure 3.

TABLE III. Type III, class 1, flat, end cutting, carbide tipped

Tip length nominal A	Thickness ± 0.010 B	Shank length nominal C	Width ± 1/32 D	End radius nominal E	Angle (maximum permissible) F	Handle length ± 1/2	Handle diameter ± 1/8
Inch	Inch	Inches	Inch	Inches	Degrees	Inches	Inches
1/4	1/8	9-1/4	1	20	3	4-3/4	1-9/16

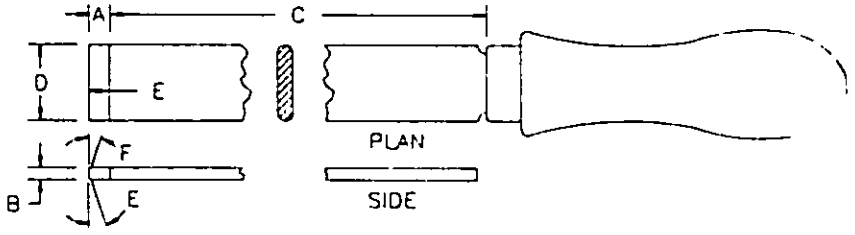


Figure 3. Type III, class 1, flat, end cutting, carbide tip.

3.8.2 Class 2, flat, end cutting, steel. Class 2 scrapers shall be flat and shall have not less than 1/64 inch radius on the corners. Class 2 scrapers shall conform to table IV and be similar to figure 4.

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TABLE IV. Type III, class 2, flat, end cutting, steel

Blade length ± 1/2	Blade thickness ± 0.010	Blade width ± 1/32	Handle length ± 1/2	Handle diameter ± 1/8
Inches	Inch	Inch	Inches	Inches
5	1/16	3/8	4-1/2	1-1/4
8	1/8	1	4-3/4	1-9/16

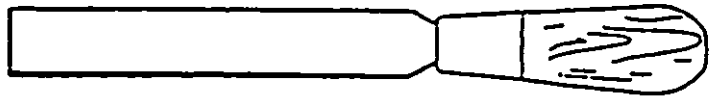


Figure 4. Type III, class 2, flat, end cutting, steel.

3.9 Type IV, double end, steel. Type IV scrapers shall be forged from either 1/2 inch diameter, round, or 1/2 inch across the flats, steel. Both ends shall taper approximately three inches to form the scraping edges. One end shall be ground square to the horizontal axis. The opposite end shall be convex. Type IV scrapers shall conform to table V and be similar to figure 5.

TABLE V. Type IV, flat, double end, steel

Width of square end ± 1/16	Width of convex end ± 1/16	Radius of convex surface ± 1/2
Inch	Inches	Inches
7/8	1-1/8	2

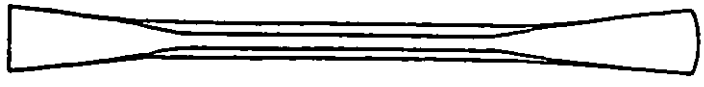


Figure 5. Type IV, flat, double end, steel.

3.10 Workmanship. Scrapers shall be free of pits, seams, laps, heat checks, scale, excessive roughness, or any other defect that will impair its usefulness.

3.11 Metric product. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this Federal Specification are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specification and standards to the extent specified, or, if none, in accordance with this specification.

4.2 Sampling procedures. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.3 Visual and dimensional examination. The end items shall be examined to determine conformance to the visual and dimensional requirements of this specification. The lot shall consist of all scrapers of the same type or class presented at one time. The sample unit shall be one scraper. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.5.

4.4 Test Methods. The end items shall be tested for the characteristics specified in 3.3. The lot size shall be expressed in units of scrapers. The sample unit shall be one scraper. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.1 Hardness. The Rockwell hardness tests shall be conducted in accordance with ASTM E 18. Care shall be exercised to assure that any decarburized surfaces are removed prior to the test.

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table VI. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

TABLE VI. Preparation for delivery defects

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling Distortion of container
Contents	Number per container is more or less than required Net weight exceeds requirements

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be A, B, or Commercial, as specified (see 6.1).

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type, class, and style required (see 1.2).
c. Blade length and width required (see applicable tables).
d. Whether scrapers should be furnished without handles (see 3.2.1).
e. Level of preservation, packaging, and packing required (see 5.1).

MILITARY INTERESTS:

Military Coordinating Activity

Army - GL

Custodians

Army - GL
Air Force - 99

Review Activity

Air Force - 84

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

GSA - FSS

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